

Jasper and Stella: distillation of SOTA embedding models

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Abstract

A crucial component in many deep learning applications, such as Frequently Asked Questions (FAQ) and Retrieval-Augmented Generation (RAG), is dense retrieval. In this process, embedding models transform raw text into numerical vectors. However, the embedding models that currently excel on text embedding benchmarks, like the Massive Text Embedding Benchmark (MTEB), often have numerous parameters and high vector dimensionality. This poses challenges for their application in real-world scenarios. To address this issue, we propose a **novel multi-stage distillation framework** that enables a smaller student embedding model to distill multiple larger teacher embedding models through three carefully designed losses. Meanwhile, we utilize Matryoshka Representation Learning (MRL) to reduce the vector dimensionality of the student embedding model effectively. Our student model named Jasper with 2 billion parameters, built upon the Stella embedding model, obtained the No.3 position on the MTEB leaderboard (as of December 24, 2024), achieving average 71.54 score across 56 datasets. We have released the model and data on the Hugging Face Hub ^{1 2}, and the training codes are available in this project repository ³.

1 Introduction

With the rapid development of natural language processing technologies, text embedding models play a crucial role in text representation (Kashyap et al., 2024), information retrieval (Zhao et al., 2024a), and text generation tasks (Gao et al., 2023). By mapping words, sentences, or documents into a

high-dimensional continuous space, these models bring similar texts closer together in their vector representations, thereby not only enhancing the manipulability of textual data but also significantly improving the performance of various downstream tasks (Agarwal et al., 2024; Wang et al., 2024; Zhou et al., 2024).

However, embedding models that demonstrate excellent performance on the METB leaderboard⁴ (Muennighoff et al., 2023) usually contain a large number of parameters and high vector dimensions. For instance, both NV-Embed-v2 (Lee et al., 2024; Moreira et al., 2024) and bge-en-icl (Xiao et al., 2023; Li et al., 2024) have 7 billion parameters and 4096-dimensional vector representations. These characteristics lead to slow inference speeds and high storage costs, posing a significant challenge to their direct practical application.

To address the aforementioned challenges, we propose a novel multi-stage knowledge distillation framework for embedding models. **Knowledge distillation** is widely recognized for enhancing the effectiveness of dense retrieval training (Hofstätter et al., 2021; Lin et al., 2021). In our framework, we introduce three carefully designed loss functions to distill knowledge from the teacher model to the student model. These loss functions shift from a specific constraint to a broader constraint. The first, **cosine loss**, calculates the absolute difference in text representations between the student and teacher models. The pointwise signal derived from a single text is straightforward, yet its limited optimization direction tends to readily lead to overfitting on the training data. Thus, we introduce the **similarity loss**, which measures the semantic discrepancies between the student and teacher models from a text-pair perspective. Additionally, we design the **relative similarity distillation loss** to further leverage relative ranking information. This

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¹https://huggingface.co/infgrad/jasper_en_vision_language_v1

²https://huggingface.co/datasets/infgrad/jasper_text_distill_dataset

³<https://github.com/NLPJCL/RAG-Retrieval>

⁴<https://huggingface.co/spaces/mteb/leaderboard>

ensures that the student model learns the teacher’s ranking preferences across all potential positive and negative text pairs within the batch, thereby improving the robustness of embedding learning.

To further improve the performance of the student model, we utilize **multiple powerful** large embedding models as teachers. Specifically, we concatenate the vectors produced by all teacher models to create the final ground truth, which inevitably leads to an increase in the student model’s vector dimension. To address this issue, we adopt a Matryoshka Representation Learning (MRL)-based training method (Kusupati et al., 2024) to effectively compress the student model’s vector representation. Additionally, to develop the multi-modal retrieval capability of our student model, we integrate a vision encoder and introduce a self-distillation mechanism to align the visual embeddings with the textual embeddings. In terms of the overall training process, we employ a 4-stage distillation approach to progressively transfer knowledge from the teacher models to the student model. Each stage focuses on specific aspects, combining **three loss functions and fine-tuning different parameters** of the student model to ensure a smooth and effective distillation process.

Experimental results on the MTEB leaderboard demonstrate that our student model named Jasper with 2 billion (2B) parameters, primarily built upon the foundation of the Stella embedding model, delivers excellent performance (average 71.54 score across 56 datasets) comparable to other embedding models with 7 billion (7B) parameters, and significantly outperforms models with fewer than 2B parameters.

The main contributions of this paper can be summarized as follows:

- (1) We propose a novel multi-stage distillation framework, which enables a smaller student embedding model to effectively distill knowledge from multiple larger teacher embedding models through three carefully designed loss functions.
- (2) Our 2B Jasper model obtained the No.3 position on the MTEB leaderboard (as of December 24, 2024), producing results comparable to other top-ranked 7B embedding models and significantly outperforming other models with less than 2B parameters.

2 Methods

2.1 Definitions

For a more comprehensive introduction of our model and distillation framework, we make the following definitions:

- Student Model: The text embedding model that is the subject of training, tasked with learning to produce effective vector representations.
- Teacher Model: The state-of-the-art (SOTA) embedding model serving as a teacher, guiding the student model in generating effective vectors. Notably, the teacher model will not be trained.
- s_x : The normalized vector representation of a text x produced by the student model.
- t_x : The vector representation of the same text x , first normalized, then concatenated, and normalized again, produced by multiple teacher models.
- S_X : A matrix of normalized vector representations for a batch of text X produced by the student model.
- T_X : A corresponding matrix of vector representations for the same batch of text X , first normalized, then concatenated, and subsequently normalized again, generated by multiple teacher models.

2.2 Model Architecture

Our student model architecture follows the simple and standard design of combining a language model with a vision encoder. As shown in Figure 1, it consists of four components:

1. A encoder-based language model that generates text embeddings through mean pooling.
2. A vision encoder that independently maps images into vision token embeddings.
3. A pooler that maps vision token embeddings to the same dimension as the language model’s input textual embeddings, while reducing the length of visual token sequences.
4. Several fully connected (FC) layers that project the embeddings to a specific dimension for the final output.

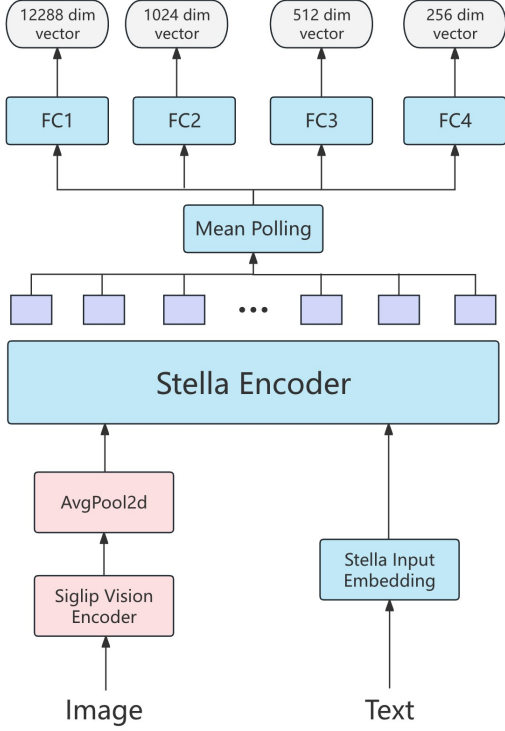


Figure 1: The model architecture of Jasper model.

2.3 Stage 1&2: Distillation from Multiple Teachers

In the first two stages of distillation, we use a fully connected layer to map the vectors of the student model onto the dimensions of the teacher models. Specifically, we employ *NV-Embed-v2*⁵ and *stella_en_1.5B_v5*⁶ as teacher models, which have vector dimensions of 4096 and 8192, respectively. After the mapping process, the student model’s vector dimension is adjusted to 12288, equal to the combined vector dimensions of two teacher models (4096 + 8192).

The objective of the first two stages is to enable the student model to effectively learn text representations from multiple teacher models by aligning its output vectors with the corresponding teacher vectors. To achieve this goal, we carefully design three loss functions that progress from a specific to a broader perspective. The first loss function is cosine loss, which is formulated as follows:

$$\mathcal{L}_{\cosine} = \sum_x 1 - s_x \cdot t_x. \quad (1)$$

⁵<https://huggingface.co/nvidia/NV-Embed-v2>

⁶https://huggingface.co/dunzhang/stella_en_1.5B_v5

The $\mathcal{L}_{\cosine}$ is designed to minimize the angular difference between student and teacher vectors in the high-dimensional space, with the aim of aligning their absolute text representations. However, the $\mathcal{L}_{\cosine}$ value generally does not converge to zero, suggesting a persistent angular discrepancy between the student and the teachers. Meanwhile, the pointwise signal derived from a single text has a limited optimization direction, which can easily lead to overfitting on the training data.

$$\mathcal{L}_{sim} = MSE(S_X S_X^T, T_X T_X^T) \quad (2)$$

To complement the limitations of $\mathcal{L}_{\cosine}$, we introduce the second loss function, similarity loss, as defined in (2), which models the semantic matching differences between the student and teacher models from a text-pair perspective. This loss function ensures a relatively consistent judgment of similarity between the student model and the teacher models, without enforcing an absolute fit between the student model and the teacher model.

$$\mathcal{L}_{resim} = \frac{1}{N} \sum_{t_i \cdot t_j > t_m \cdot t_n} MAX(0, s_m \cdot s_n - s_i \cdot s_j + margin) \quad (3)$$

To further leverage relative comparison signals, inspired by CoSENT loss⁷, we propose the third loss function, relative similarity distillation loss, as defined in (3). For each batch of text data, we employ teacher models to automatically generate soft labels for all text pairs, thereby identifying potential positive and negative samples. Subsequently, the student model is trained to ensure that the similarity between positive pairs exceeds that between negative pairs, with the *margin* hyperparameter controlling the degree of this difference. If the batch size is m , the total number of text pairs (*i.e.*, N) is given by C_m^2 .

$$\mathcal{L} = \lambda_1 \mathcal{L}_{\cosine} + \lambda_2 \mathcal{L}_{sim} + \lambda_3 \mathcal{L}_{resim} \quad (4)$$

The final loss $\mathcal{L}$ is a weighted sum of the aforementioned three loss functions. where λ_1, λ_2 , and λ_3 are hyperparameters. The biggest advantage of distillation vectors is that we do not need any supervised data. Without considering resource constraints, we can use **trillions of unsupervised texts**

⁷<https://spaces.ac.cn/archives/8847>

for distillation training to achieve extreme performance for a given model size.

Notably, the main difference between stage 1 and stage 2 lies in the trained parameters. In stage 1, only the fully connected layer (FC1) is trained, whereas in stage 2, both the fully connected layer (FC1) and the last three encoder layers of the student model are trained.

2.4 Stage 3: Dimension Reduction

In the first two stages, the student model is trained by learning from the teacher models. Specifically, we concatenate the vectors produced by the two teacher models, resulting in a student model vector with a dimensionality of 12,288 (4,096 + 8,192), which is impractically large. Inspired by MRL (Kusupati et al., 2024), we introduce three additional, independent fully connected layers (FC2, FC3, and FC4) to generate low-dimensionality vectors, each achieving a different level of dimension reduction. For instance, by incorporating the fully connected layer FC3 with a shape of (1536⁸, 512), we obtain a more manageable 512-dimensional vector space.

For the three FC layers, since the dimensions of the reduced vectors do not align with those of the concatenated teacher vector, the $\mathcal{L}_{\cosine}$ is omitted and only the $\mathcal{L}_{sim}$ and $\mathcal{L}_{resim}$ are utilized. To ensure the accuracy of the vectors generated from the FC1 layer (i.e., the 12288-dimensional vectors), they continue to be trained using all three loss functions. During this stage, all parameters of the student model are trained.

In addition to the previously mentioned dimension reduction method, we present a potentially promising approach to self-distillation, where the aligned vectors from an earlier stage of the student model’s training serve as teacher vectors. Specifically, we propose to utilize the 12288-dimensional vectors output from the FC1 layer to serve as teachers for the shorter vectors generated by the other three FC layers. This approach provides a unique advantage by enabling the reduction of the dimensionality of any embedding model, utilizing only unsupervised data and the model itself. Given that this paper primarily focuses on introducing the training methods of the Stella and Jasper models, we did not conduct experiments to evaluate the specific merits of this proposed approach.

⁸This refers to the dimensionality of the encoder layer’s hidden state.

2.5 Stage 4: Unlock Multimodal Potential

In stage 4, we leverage image-caption pairs as the training dataset, focusing exclusively on training the visual encoder while keeping the other components frozen. The training process is based on self-distillation, where the caption’s vector representation serves as the teacher vector, and the image’s vector representation acts as the student vector. All fully connected layers introduced in previous stages are employed to generate multiple pairs of student and teacher vectors. For each pair, we calculate three losses, which are then averaged to obtain the final loss.

It is important to note that this stage achieves only a preliminary alignment between the text and image modalities, leaving significant room for improvement. In future work, we aim to further explore and refine the modality alignment process.

3 Experiments

3.1 Implementation details

Our model named Jasper is initialized from *stella_en_1.5B_v5* and *google/siglip-so400m_patch14-384* (Zhai et al., 2023; Alabdulmohsin et al., 2024). *stella_en_1.5B_v5* and *NV-Embed-v2* are our teacher models. The total number of parameters in our Jasper model is 1.9B (*stella* 1.5B parameters and *siglip* 400M parameters). For hyperparameters, we set $\lambda_1 = 10$, $\lambda_2 = 200$, $\lambda_3 = 20$, margin = 0.015.

In all four stages, the model is trained using 8 × RTX A6000 GPUs, with a maximum input length of 512 tokens, mixed precision training (BF16), DeepSpeed ZERO-stage-2, and the AdamW optimizer. During stage 1 (distillation training), the batch size is set to 128, the learning rate is 1e-4 per step, and the model checkpoint at step 4000 is selected as the final model. In the case of stage 2 (also distillation training), the batch size remains 128, the learning rate drops to 8e-5 per step, and the final model is the checkpoint at step 7000. For stage 3 (dimension reduction training), the batch size is again 128, the learning rate is adjusted to 7e-5 per step, and the checkpoint at step 2200 serves as the final model. Lastly, in stage 4 (multimodal training), the batch size is reduced to 90, the learning rate returns to 1e-4 per step, and the final model is chosen from the checkpoint at step 3500.

Model	Model Size	Average(56 datasets)	Classification	Clustering	PairClassification	Reranking	Retrieval	STS	Summarization
NV-Embed-v2	7851M	72.31	90.37	58.46	88.67	60.65	62.65	84.31	30.7
bge-en-icl	7111M	71.67	88.95	57.89	88.14	59.86	62.16	84.24	30.77
Stella_en_1.5B_v5	1543M	71.19	87.63	57.69	88.07	61.21	61.01	84.51	31.49
SFR-Embedding-2_R	7111M	70.31	89.05	56.17	88.07	60.14	60.18	81.26	30.71
gte-Qwen2-1.5B-instruct	1776M	67.16	82.47	48.75	87.51	59.98	58.29	82.73	31.17
voyage-lite-02-instruct	1220M	67.13	79.25	52.42	86.87	58.24	56.60	85.79	31.01
Jasper (our model)	1543M+400M	71.54	88.49	58.04	88.07	60.91	61.33	84.67	31.42

Table 1: MTEB Results as of December 24, 2024. We use the original model names on the leaderboard for clarity.

3.2 Datasets

In stage 1, stage 2 and stage 3, we use *fineweb-edu* (Lozhkov et al., 2024) as our main text training dataset, which accounts for 80% of the full text data. The remaining 20% of the text data comes from *sentence-transformers/embedding-training-data*⁹. The reason we choose the *sentence-transformers/embedding-training-data* is that the majority of the *fineweb-edu* data consists of passages. However, in addition to passages, we also require questions to enhance the diversity of our training data. The total amount of text training data is 8 million.

For the documents in our dataset, we perform the following actions:

1. We randomly select 30% of the documents and divide them into short texts, each consisting of 1 to 10 sentences.
2. We randomly select 0.08% of the text and shuffle the words within it.

In stage 4, we use the caption data of *BAAI/Infinity-MM* (Gu et al., 2024) as our vision training data.

3.3 Results

We evaluate the proposed Jasper and Stella models on the full MTEB benchmark, which encompasses 15 retrieval datasets, 4 reranking datasets, 12 classification datasets, 11 clustering datasets, 3 pair classification datasets, 10 semantic textual similarity datasets, and 1 summarization dataset.

Table 1 presents the average score of our Jasper model across the overall performance and seven subcategory tasks of the METB benchmark. We compare our model with other frontier models on the MTEB leaderboard, as well as those with fewer than 2B parameters. Experimental results demonstrate that our Jasper model significantly outperforms other models with fewer than 2B parameters.

⁹<https://huggingface.co/datasets/sentence-transformers/embedding-training-data>

Furthermore, despite having only 2B parameters, our model produces results that are comparable to those of models with 7B parameters.

4 Discussion

4.1 Instruction Robustness

Instruction-based embedding models require an instruction to be prepended to a query or passage during text encoding. Currently, many state-of-the-art text embedding models use instructions to prompt the model and obtain better embeddings. Similar to the usage of large language models (Zhao et al., 2024b), different tasks necessitate different instructions, which is both logical and intuitive. Therefore, the ability to understand instructions is crucial for these text embedding models.

Jasper is also an instruction-based embedding model. To demonstrate the impact of different prompts on the Jasper model, we conducted a simple experiment. Specifically, we evaluated Jasper on some short evaluation tasks using similar instructions generated by GPT-4o. Table 2 lists all the original and modified instructions. Based on the results shown in Table 3, we conclude that our Jasper model is robust to instructions and can accurately understand different instructions.

4.2 Possible Improvements for Vision Encoding

Due to time and resource constraints, we were only able to equip the Jasper model with a basic image encoding capability. Initially, stage 4 was envisioned as a fundamental visual-language alignment training phase, with a potential stage 5 involving contrastive learning utilizing a Visual Question Answering (VQA) dataset. Additionally, we observed oscillatory behavior in our loss function during stage 4. Overall, there is considerable room for enhancement in the multimodal training.

5 Conclusion

In this paper, we present the distillation-based training procedure for the Jasper model. We have

Original Instruction	Synonym of Original Instruction
Classify the sentiment expressed in the given movie review text from the IMDB dataset	Determine the sentiment conveyed in the provided movie review text from the IMDB dataset.
Identify the topic or theme of StackExchange posts based on the titles	Determine the subject or theme of StackExchange posts based on the titles.
Given a news summary, retrieve other semantically similar summaries	Given a news summary, find other summaries with similar meanings.
Retrieve duplicate questions from StackOverflow forum	Find duplicate questions on the StackOverflow forum.
Given a title of a scientific paper, retrieve the titles of other relevant papers	Given the title of a scientific paper, find the titles of other related papers.
Classify the sentiment of a given tweet as either positive, negative, or neutral	Determine the sentiment of a given tweet as positive, negative, or neutral.
Given a claim, find documents that refute the claim	Given a claim, locate documents that contradict the claim.
Given a question, retrieve relevant documents that best answer the question	Given a question, find relevant documents that best answer it.
Retrieve tweets that are semantically similar to the given tweet	Find tweets that have similar meanings to the given tweet.
Retrieve semantically similar text.	Find text with similar meanings.
Identify the main category of Medrxiv papers based on the titles	Determine the primary category of Medrxiv papers based on the titles.
Retrieve duplicate questions from AskUbuntu forum	Find duplicate questions on the AskUbuntu forum.
Given a question, retrieve detailed question descriptions from Stackexchange that are duplicates to the given question	Given a question, find detailed question descriptions from Stackexchange that are duplicates.
Identify the main category of Biorxiv papers based on the titles and abstracts	Determine the primary category of Biorxiv papers based on the titles and abstracts.
Given a financial question, retrieve user replies that best answer the question	Given a financial question, find user replies that best answer it.
Given an online banking query, find the corresponding intents	Given an online banking query, identify the corresponding intents.
Identify the topic or theme of the given news articles	Determine the subject or theme of the given news articles.
Classify the emotion expressed in the given Twitter message into one of the six emotions: anger, fear, joy, love, sadness, and surprise	Determine the emotion expressed in the given Twitter message as one of six emotions: anger, fear, joy, love, sadness, and surprise.
Given a user utterance as a query, find the user intents	Given a user utterance as a query, identify the user intents.
Identify the main category of Biorxiv papers based on the titles	Determine the primary category of Biorxiv papers based on the titles.
Classify the given Amazon review into its appropriate rating category	Classify the given Amazon review into its appropriate rating category.
Given a scientific claim, retrieve documents that support or refute the claim	Given a scientific claim, find documents that support or contradict the claim.
Identify the topic or theme of StackExchange posts based on the given paragraphs	Determine the subject or theme of StackExchange posts based on the given paragraphs.
Given a scientific paper title, retrieve paper abstracts that are cited by the given paper	Given a scientific paper title, find paper abstracts that are cited by the given paper.
Classify the given comments as either toxic or not toxic	Classify the given comments as toxic or non-toxic.
Classify the intent domain of the given utterance in task-oriented conversation	Determine the intent domain of the given utterance in task-oriented conversation.
Retrieve duplicate questions from Sprint forum	Find duplicate questions on the Sprint forum.
Given a user utterance as a query, find the user scenarios	Given a user utterance as a query, identify the user scenarios.
Classify the intent of the given utterance in task-oriented conversation	Determine the intent of the given utterance in task-oriented conversation.
Classify a given Amazon customer review text as either counterfactual or not-counterfactual	Classify a given Amazon customer review text as counterfactual or non-counterfactual.
Identify the main category of Medrxiv papers based on the titles and abstracts	Determine the primary category of Medrxiv papers based on the titles and abstracts.
Given a query on COVID-19, retrieve documents that answer the query	Given a query on COVID-19, find documents that answer the query.

Table 2: Original instructions and corresponding synonyms.

Task Type	Task Name	Original Score	Score with Modified Instructions
Classification	MTOPIDomainClassification	0.992	0.992
Classification	AmazonCounterfactualClassification	0.958	0.957
Classification	TwetSentimentExtractionClassification	0.773	0.776
Classification	EmotionClassification	0.877	0.859
Classification	MassiveIntentClassification	0.853	0.854
Classification	AmazonReviewClassification	0.629	0.630
Classification	MassiveScenarioClassification	0.912	0.912
Classification	Banking77Classification	0.873	0.875
Classification	ImdbClassification	0.971	0.971
Classification	ToxicConversationsClassification	0.913	0.910
Classification	MTOPIIntentClassification	0.915	0.912
Clustering	MedrxivClusteringS2S	0.448	0.448
Clustering	StackExchangeClusteringP2P	0.494	0.492
Clustering	StackExchangeClustering	0.800	0.795
Clustering	TwentyNewsgroupsClustering	0.630	0.625
Clustering	MedrxivClusteringP2P	0.470	0.468
Clustering	BiorxivClusteringS2S	0.476	0.475
Clustering	BiorxivClusteringP2P	0.520	0.518
PairClassification	TwitterURLCorpus	0.877	0.877
PairClassification	SprintDuplicateQuestions	0.964	0.964
PairClassification	TwitterSentEval2015	0.803	0.801
Reranking	StackOverflowDupQuestions	0.546	0.548
Reranking	SciDocRR	0.891	0.890
Reranking	AskUbuntuDupQuestions	0.674	0.676
Retrieval	CQA4DupstackMathematicaRetrieval	0.369	0.370
Retrieval	CQA4DupstackStateRetrieval	0.413	0.413
Retrieval	CQA4DupstackTextRetrieval	0.362	0.362
Retrieval	SciDOCS	0.247	0.247
Retrieval	CQA4DupstackEnglishRetrieval	0.543	0.543
Retrieval	ArgoAns	0.653	0.652
Retrieval	TRECCOVID	0.865	0.866
Retrieval	CQA4DupstackUnixRetrieval	0.482	0.482
Retrieval	CQA4DupstackGamingRetrieval	0.632	0.633
Retrieval	CQA4DupstackGisRetrieval	0.444	0.448
Retrieval	CQA4DupstackWordpressRetrieval	0.388	0.386
Retrieval	FiQA2018	0.601	0.601
Retrieval	SciFact	0.805	0.805
Retrieval	CQA4DupstackPhysicsRetrieval	0.549	0.548
Retrieval	NFCorpus	0.431	0.431
Retrieval	CQA4DupstackProgrammersRetrieval	0.505	0.505
Retrieval	CQA4DupstackAndroidRetrieval	0.571	0.571
Retrieval	CQA4DupstackWebmastersRetrieval	0.464	0.464
STS	BIOSES	0.848	0.854
STS	STS13	0.897	0.888
STS	STS12	0.803	0.804
STS	STS1Benchmark	0.888	0.886
STS	STS15	0.902	0.900
STS	STS14	0.853	0.851
STS	STS16	0.864	0.869
STS	STS22	0.672	0.748
STS	SICK-R	0.822	0.823
STS	STS17	0.911	0.908
Summarization	SummEval	0.313	0.314
Average Score		0.686	0.687

Table 3: MTEB Results on different instructions.

designed three loss functions to distill multiple large teacher embedding models into a student embedding model from diverse perspectives. Subsequently, we utilized a MRL-based training method to reduce the vector dimensionality of the student model. Experimental results on the MTEB demonstrate that our Jasper model achieves state-of-the-art performance at the 2B parameter scale and exhibits comparable results to other top-ranked embedding models with 7B parameters. Future work will further explore the alignment between multiple modalities.

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